# CORE:HACK

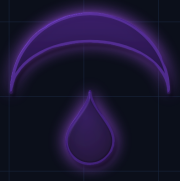
## BLACKOUT

### ADVANCED MANUAL

A field guide to the operator's programs

*Those who know their tools*
*leave fewer traces.*



Core:Hack: Blackout 2.10.2 - Guido Cauli <guido.cauli@gmail.com>

01

# The Operator's trade

A program is a tool, not magic. Whoever taught you the trade always said so: the difference between an operator and a target lies in what they do in the ten seconds before pressing Enter.

This guide goes through the programs you will use every day, in the order you usually need them, with their general use and a few habits picked up in the field. It is not a list of secrets: you will find the secrets on your own, and some of them you would rather not find.

## Level, noise and speed

Every protection has a level, and so does every program. There is only one rule: **the program must be worth at least as much as the defence**. One or two levels short, you can still try, but you work more slowly and the version is reported as not recognised; three short, you do not get through. Extra levels do the opposite: you work faster and make less noise. Noise is what the Trace listens to.

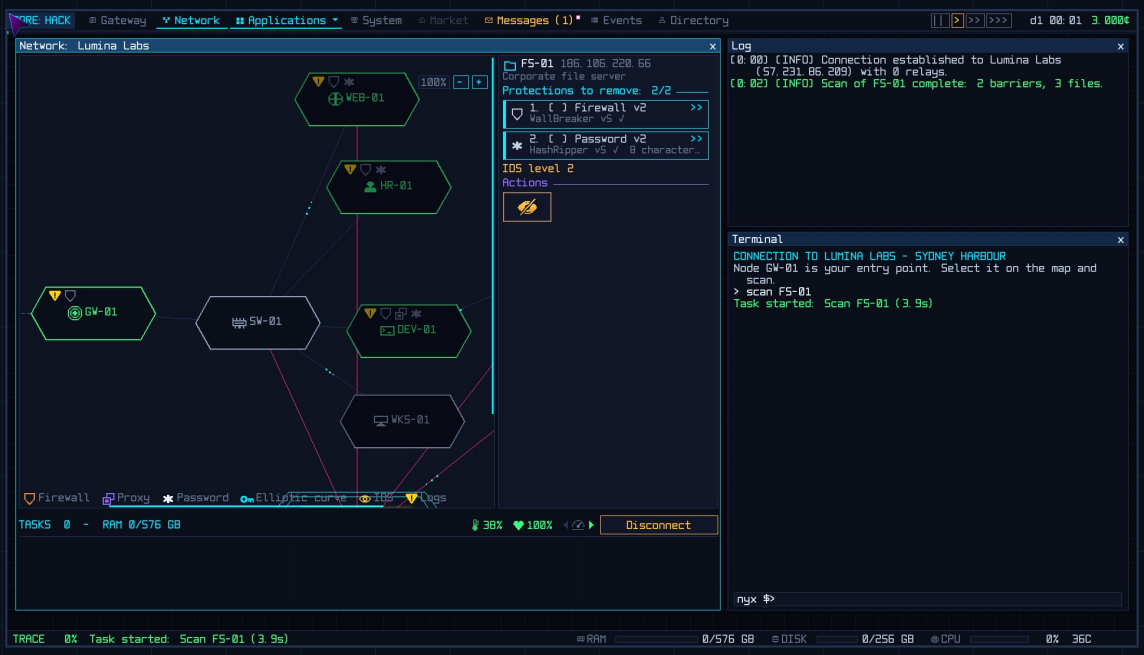
| What you look at | What it tells you |
|---|---|
| **Program level** | Which defences you are useful against, and how fast. |
| **Memory slots** | How many programs run together. Each one holds a slot until it finishes. |
| **Processor speed** | How long tasks take. More tasks in parallel share the same power. |
| **Trace** | How close you are to being found. It rises with noise, and it does not cool down by itself. |

The processor can be pushed past its normal rating with the `oc` command: more speed right away, but more heat, and a cooked processor has to be repaired. Use it in bursts, not out of habit.

## The typical workflow

Almost every intrusion follows the same rhythm, even when the details change:

| Phase | What you do |
|---|---|
| 1 Look | Scan the node and read what it shows you: defences, files, any watchers. |
| 2 Prepare | If there is a monitor, blind it. Pick your programs and free the slots you need. |
| 3 Open | Break the defences one by one, or in groups if memory allows. |
| 4 Take | Decrypt, download, alter or delete what the contract needs, nothing more. |
| 5 Clean | Wipe the logs of the nodes you touched, then leave. |




*An intrusion session: node map, running tasks, log and terminal.*

02

# Reconnaissance

You do not open a door without knowing where it leads. The programs in this chapter break nothing: they tell you what is there.

## NetSonar

*Node scanning*

It is the first program you run on any host. A scan reveals the protections in place, their level, the files and whether the node is watched. Until you have scanned it, the node is a closed box. On a host that is already open a scan matters less, but it is still useful to know whether anything has changed.

**From the field:** Always scan before choosing your programs: bringing the wrong level costs minutes the Trace will not give back. A higher-level NetSonar is also faster and more discreet.

## ArpScan

*Listening to nearby traffic*

From a host that is already open it listens to local traffic and draws the neighbouring nodes and their defences on the map, up to two hops (three with the best version). It does not replace scanning a single node, but it saves a lot of ground when the network is wide.

**From the field:** Run it as soon as you have a solid foothold in the middle of the network: it shows you where it pays to go, and which branches to leave alone.

## Inspect

*Anomalous digital signatures*

Some files carry something the eye cannot see: a hidden mark, or a nature someone has an interest in concealing. Inspect reads anomalous signatures. On an open host you run it with `inspect <node>` (fast, low noise): the files it recognises turn violet and the log warns you. On your own disk it works by itself and checks every file that arrives.

**From the field:** It only recognises what its level can read. A file that does not change colour is not clean because of it: something subtler may slip past. Whoever downloads blindly brings other people's problems home.

## LogParser

*From logs to gateway*

The `.log` files on hosts keep the traces of whoever passed through. With `logparse <node>` LogParser reads them and follows the relay chain of an operator on the ladder: each version reveals one more address, and when the chain is complete you reach the gateway. You get a dossier, and the operator appears in the Directory with their card and the door to their private network. Requires NetSonar v2 and GhostShell v2.

**From the field:** The higher an operator's grade, the more relays they have: a long chain needs a high-version LogParser. If the operator gets arrested everything changes, and the dossier is worth nothing any more.

## FileUndeleter

*Deleted files*

Some logs have already been deleted by someone else when you arrive, and those are exactly the ones that talk about an operator. `undelete <node>` recovers them, up to the version the program can read (1-3, depending on difficulty). Requires GhostShell v2 and Inspect v1.

**From the field:** Recover first, then read: LogParser only works on what it finds. And remember that rummaging leaves traces of you too.

## XMapper

*Secret links and hosts*

From grade-8 contracts upward some networks hide shortcuts between distant hosts, and entire hosts nobody told you to look for. XMapper is a NetSonar addon: once installed, the normal scan of a host reveals its secret links (dashed in violet) and the hidden hosts they lead to. A single level, expensive, sold only at faction gateways; requires NetSonar v4 and Inspect v2. Argus already includes it.

**From the field:** A secret host is an ordinary host, with its own defences: hidden, not harder. It holds loot files, in orange, that pay an extra fee when the contract succeeds. They are not part of the contract's trail: if you do not look for them, the contract stays the same.

03

# Breaking the defences

Four programs, four kinds of protection. This is where it starts, and these are the ones that teach you the rhythm.

| Defence | Broken with | How |
|---|---|---|
| Firewall | WallBreaker | disables it |
| Access proxy | ProxyMask | bypasses it |
| Password | HashRipper | cracks it |
| Elliptic curve | CurveBreaker | forces it |

### WallBreaker

*Firewall*

A node behind a firewall will not talk to you until you disable it. WallBreaker works on the chosen node with `break <node>` and stops on its own when done. It does nothing else.

**From the field:** If two defences block the same node and you have free memory, run both programs together: the power is shared, but the times overlap.

### ProxyMask

*Access proxy*

Many nodes cannot be reached directly: they sit behind a proxy that filters who may pass. ProxyMask gets around it, giving your traffic an acceptable face.

### HashRipper

*Password*

Passwords are cracked with patience. Once found, the password stays noted for the duration of the contract: if it is still valid you can get back in without repeating the work and without the Trace noticing. On the node the mark of a known password blinks green. Node passwords change over time, but never while you are connected.

**From the field:** A password you cracked and that has since changed is worth nothing: if you come back hours later, expect to do the work again.

### CurveBreaker

*Elliptic-curve cryptography*

Elliptic-curve keys are the hardest locks to force in a network. CurveBreaker works for a long time and eats power: before you run it, free the slots and check the level.

**From the field:** Do not start a CurveBreaker halfway through a session if your Trace is high. Better to decide beforehand whether it is worth it.

## Going where nobody is looking

When you break a defence, its symbol on the node turns grey. Only when all the protections have fallen does the node open, and only then can you touch the files. If you lack the right program, the matching button does not appear: its absence already tells you how prepared you are.

04

# Files

### CipherKey

*Decryption*

An encrypted file cannot be downloaded, deleted or altered until you have decrypted it. You run it with `decrypt <node> <n>`. Once open, the file is treated like any other: `get` to download it, `rm` to delete it, `edit` to alter it.

**From the field:** Downloading is not instant: it depends on the network card. A large node with heavy files means minutes in which the Trace climbs.

### DBEdit

*Database tables*

Some hosts keep tables (`.dbt` files): grades, medical records, civil records, balances, criminal records. Without DBEdit they are files like any other. With DBEdit you can read and edit them: `db show`, `db alter`, `db delete`. On remote nodes every change is a task like any other; on your own copies it is instant and leaves no marks.

**From the field:** Deleting a record only hides it: it vanishes from the table and the rows are renumbered, but someone with the right tools can bring it back.

### DBDiag and DBRebuild

*Diagnosis and restore*

`db diag` tells you whether a table holds hidden records, and how many. `db rebuild` brings them back into view. You need them when someone has tampered with a table and the client wants the original data.

**From the field:** Every table has a version: a program only handles versions equal to or lower than its own.

## Storage

Downloaded files stay on your disk until you remove them, and space is not infinite. From the Storage tab of the System window (or with `drop`) you delete them. Keep only what you need: a full disk stops you halfway through a download, and some things on your disk are more dangerous to keep than not to have.

05

# Staying invisible

Every action makes noise. These are the tools to keep it low, or to make whoever is listening look elsewhere.

### GhostShell

*IDS monitors and logs*

It does two things. With `blind <node>` it blinds the monitor of a watched node for a while: as long as it lasts, your moves go unnoticed. With `wipe <node>` it erases the logs you left behind. It is expensive, but it is the tool that separates a clean job from one that comes looking for you.

**From the field:** Blinding does not last long. If the monitor comes back on while you work, the Trace rises faster. Launch everything else only afterwards, and keep an eye on the time left.

### TraceVeil

*A veil over your traffic*

A resident program: you start it with `veil` and stop it the same way. While it is active it lowers the noise of your tasks and hides part of the monitors. At high levels it weighs on memory like any other program.

**From the field:** The Trace sees through veils of its own level: against heavily watched networks a low veil is of little use.

## Relays and Cloak

A **relay** is an intermediate host between you and the target: each bounce slows the Trace, but adds latency. How many you can chain and how much they are worth depends on your **Cloak** component. You see the relays you have with `relays`: the one from a friend in the trade, the ones you planted yourself and the ones you lease for a time with `relay lease`. Before each connection you choose how many to use, and the number you chose is waiting for you when you return.

A burnt relay does not come back: if you are caught, you lose it. Use them sparingly.

### DeepRoot

*Rootkits and permanent relays*

On an open dormant host, `root <node>` plants a rootkit that turns it into one of your relays. It stays yours even after the contract. Better versions unlock extra modules.

**From the field:** A relay that holds only as long as nobody looks for it: if a defender finds it, you lose it, and some trust with it.

06

# Sabotage

Some contracts do not ask you to take, but to break. Those are the ones that weigh most on your conscience: choose carefully.

### ByteRot

*Malware: corrupting a system*

With access to the node, `corrupt <node>` uploads and runs the malware. The system hit is marked as corrupted and the mark appears on the node. Some nodes have endpoint protection: the malware must be at least at its level.

### LockJaw

*Cryptolocker: locking a system*

ByteRot's cousin, it works to lock rather than destroy: `lock <node>`. The same rules apply, endpoint protection included.

### TimeBomb

*Timed corruption*

With `bomb <node>` you plant a dormant corruption in an open system: it stays silent until you decide to set it off, for all armed nodes at once, with `detonate`.

**From the field:** Arming several nodes and setting everything off in one go lets you leave before anyone works out where it came from.

### Devour

*Destroying a host for good*

Requires LockJaw v3. On an open host, `devour <node>` erases it: services, files, RevShell, KeyWraith, TimeBomb and rootkit go with it, and the host vanishes from the map along with its links. It stays down even if you come back in through a RevShell. If it was the only way to other hosts, you can no longer reach them from there: look at what lies behind it before you run it. It does not work on the entry point or on the host you came back in through. It is slow, heavy and noisy, and the destruction leaves traces. Permanent companies get the host back up in three to five days, rebuilt from backups and hardened; destroying it raises your Heat a lot.

**From the field:** Its Black version, **DarkHole**, is almost silent, takes the host's logs with it and spreads to a neighbouring host: a target of the contract, if there is one, otherwise the innermost host that does not hold the contract's files.

## Every act of sabotage has a price

Whoever damages other people's systems gets noticed faster than whoever only reads. Your Heat rises, and whoever pays you to break things knows you can also be paid to do the opposite. Think about it before you accept.

07

# Beyond the visible network

## RevShell

*A door left open*

On a breached host, `revshell <node>` leaves a back door. If the mission is interrupted you can get back in once: the network keeps its state and the Trace restarts from a quarter of the value it had reached. It also works across different networks. On the permanent networks of companies, the ones you reach with free access, the RevShell stays even after you have left: from the company's card in the Directory you get back in from that host, with the network as you left it, until the administrators find it (ten days at most, at once if they trace you). It is not free, though: a RevShell still open at the end of a contract flags you to whoever is watching.

**From the field:** Close it when you no longer need it.

## LANScan and LANTraverse

*Overlaid networks*

Some networks hide another one underneath. Only **Switches** connect overlaid networks, and only Switches respond to these two programs. `lanscan <node>` on an open Switch tells you whether it is a portal and at what level you must work to cross it. `traverse <node>` takes you in, and takes you out when you repeat it from the entry Switch. They leave no logs and make little noise.

**From the field:** Every network has its own Trace. The one you leave cools down, but it does not forget: logs left on its nodes are still paid for at the end of the contract.

## KeyWraith

*Recording from the inside*

Requires DeepRoot v3 and GhostShell v3. On an already-open host of a company in the Directory, `keylog <host>` (or the node panel) plants a keystroke recorder that stays after the contract and mails you what it gathers. What it gathers depends on the host — administration leaves passwords, accounting the books and an account holder for your address book, workstations the name and access of whoever uses them, archives the way to the target — and on how deep it sits: the further in, the more it counts. Higher versions gather more often and run more recorders, but never more than one per company.

**From the field:** The Dossier shows how much it is at risk; `keylog pull <company>` pulls it out with no traces. If they find it first, the company stays on guard for a few days, and your conscience pays for it.

## Archives

At the bottom of networks sit the backup hosts (NAS, SAN, tape libraries): few barriers, huge files, and sometimes a copy of what you are deleting. Leave it there and the file comes back. In contracts that ask you to find a copy, the archive is not visible: on the host the file vanished from, look for the backup job log (`backup_jobs_*.log`) and download it. To destroy an archive, empty it first: ByteRot and TimeBomb will not start on a full archive, and a host that is down can no longer be emptied.

## Special contracts

**Unknown host.** The client knows the file but not the host: the target sits on one of the deepest hosts or on a secret host, and in networks with several VLANs not always in the last one. They are worth one grade more, plus one per VLAN. Whoever moves the files leaves a transfer log on some shallower host: download it, or read it with LogParser, and the file's host appears on the map, secret link included.

**Delivery.** The client attaches a file to the contract: once you accept the job, it lands in your storage (if there is no room, a warning, and you download it again from the contract with `attach`). It must be uploaded to the given host, from the node panel or with `put`. In the replacement variant it takes the place of a file with the same name: if that one is encrypted, decrypt it first. When the contract closes, an unused attachment disappears.

**Destruction.** One or more hosts must be erased with Devour. They appear once Devour is on the market.

## Machines at your service

The most advanced operators put together a **BotNET**: a fleet of other people's machines, used for tactical purposes. It brings no money. It serves to speed up the heaviest tasks, to blind a monitor from afar, and to prepare the ground before you go in. The more you use it, the more it gets noticed. From the terminal: `bot`, `ddos <node>`.

08

# The other side

For every operator who gets in there is someone trying to keep them out. Sooner or later it will be your turn.




*A defence session: the network to protect, the company's status and the log console.*

## TraceHound

*Tracing attackers*

The defender's companion. It reads the security logs and helps you trace suspicious addresses back. `block <ip>` stops them for a while: afterwards, the attackers change address.

**From the field:** A harmless address blocked by mistake costs the company. Do not react to everything.

## ScrubDaemon and AegisLock

*Cleaning and protection*

ScrubDaemon cleans systems infected by malware; AegisLock protects and unlocks those hit by a cryptolocker. With `scrub`, `shield` and `recover` you use them on the right nodes. Isolating and restoring nodes, or keeping an eye on some with `monitor`, completes the picture.

**From the field:** Whoever waits to see the damage before moving always arrives late. The company's uptime is your score.

## Archives under attack

A serious ransomware crew does not start with production: it starts with the backups. If the archive falls, the company will pay. On a backup host you can use `worm <host>`: for forty seconds the archive is write-once and nobody can encrypt or infect it. The price: backup jobs stop and uptime drops, and before doing it again you must wait for it to clear the backlog. Use it when the payload is already close, not out of habit.

## When you are the one under attack

Your gateway has defenders of its own: rings of programs that load when needed and take up memory like everything else. **BastionWall** holds the perimeter, **DecoyProxy** filters sessions and probes, **WatchTower** warns you before the attack lands, **CoreVault** limits what you lose. If you know in advance that someone is coming to check, choose beforehand which rings to load and at what level: memory is what you have.

09

# The bank

Real money lives in banks. Whoever can read their ledgers also knows where the trouble is, and whoever touches them answers for it.

## Statements and ledgers

*Reading and retouching transactions*

Every bank is a company in the world's Directory, with its own name: the address of its network is in the Account tab, under the bank's name. From your account you see only your own transactions. Bank networks are well defended. On the most protected host sit the account ledger (`ledger`, balances that cannot be altered: the balance is the sum of the transactions) and the statements (`stmt`) with day, payer, payee and amount. Statements are altered or deleted with DBEdit, during a mission: the real account and the running balances are recalculated.

**From the field:** A transfer has two sides. If you retouch only one, the bank notices.

## Audits

*Every 1-2 days, no discounts*

Every bank reconciles its statements with those of its counterparties. Non-existent counterparties, half-finished transactions, unauthorised accesses left on the statement: everything comes out. Hot funds seized, account frozen for two days, and a few hours later the conviction. Before the audit you must have aligned both sides or deleted the transaction that leads to you.

**From the field:** A transaction that leads to you and has not been cleaned up is a conviction waiting to happen.

## Credentials and theft

*Coming in through the front door*

Every bank has a public login: account number and password (Login tab, or `bank login <account> <password>`). An operator's number is known; the password is found by digging: a file in their gateway, the credentials tables of breached companies (if you already know their gateway or name) or a dedicated contract. Inside their account you can move any sum up to the balance, even to yourself: every operation leaves an unauthorised transaction on their statement, to be deleted in their bank's network before the audit. The next day the holder notices, changes password and investigates; and in any case every operator changes it on their own every 2-3 days: in your address book the old one stays until, when you try to log in, it turns out to be invalid. The money stays hot for 14 days and is laundered through outgoing transfers; theft weighs on your conscience.

**From the field:** Get in, move it, then clean up: statement and logs, within a day.

## Account holders

*Sixty accounts, four leaders*

Besides operators, sixty people hold an account at one of the seven banks: ordinary people, prominent figures and the four faction leaders. A person becomes a payee only after you have read their name in a ledger or a statement. Passwords: credentials tables for ordinary people and prominent figures, dedicated contracts for prominent figures and leaders (the latter only after the credentials of a prominent figure of the same faction). Leaders change password every day. A robbed account holder reports it and NetGuard investigates: if traces remain you lose reputation with their faction, and a leader sends a siege against you.

**From the field:** Those who have little weigh more on your conscience.

## Frame-ups and investigations

*The price of every move*

Contracts may ask you to **frame** an operator as the payer or payee of a dirty operation. Whoever is robbed or framed investigates you for 2-5 days: what counts is the evidence left on the statement and in the logs. From inside a bank's network, `bank inspect <name>` reads an operator's statement.

**From the field:** Without traces, the investigation runs aground.

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# Other operators online

You are not the only one working at night. On a shared server other operators live their own story alongside yours: they write to each other, pay each other, and those who accept the challenge rob each other. You can lose money and reputation, never the gateway.

## Connecting

*The server, the fingerprint, the account*

From the Applications menu open **Player network**, type the server address (`host:port`, usually port 7341) and connect. On the first connection the certificate fingerprint appears: compare it with the one given to you by whoever runs the server. The game remembers it, and if one day it changes it refuses the connection. Then you create an account or log in. From the terminal: `net connect`, `net register`, `net login`, `net status`, `net close`.

**From the field:** A fingerprint different from the expected one is not a glitch: it is a reason not to go in.

## If the line drops

*Reconnecting without losing the thread*

The game keeps the line alive with a signal every twenty seconds. If it drops, it reconnects by itself and resumes the session without asking for your password: the intrusion, race or transfer in progress carries on, and whatever was due to go out in the meantime goes out as soon as the line is back. If too much time passes, or the server has been restarted, you log back in with `net login`. Audits, password changes and server deadlines follow your clock: they do not catch you while you sleep.

**From the field:** A login to the same account from somewhere else closes this session, and does not reopen it.

## Messages and network account

*Talking and paying*

The Players tab shows who is online, with grade and faction. Messages arrive in your mail (`net msg`; `net block` silences whoever bothers you). Money between operators goes through a **network account**, separate from your own: `net bank` shows its balance, `net pay` sends a transfer, with a cap per transfer and a daily one. If the server does not confirm, the transfer waits and goes out again at the next login.

## Server players

*Not every name is a person*

Some operators on the ladder are run by the server: they have a grade, a faction and their own waking hours, and they play by everyone's rules. They can be hit, they hit back, they defend their accounts, run races and answer messages.

**From the field:** Treat them like anyone else: they remember who robbed them.

## The challenge

*Only for those who accept it*

With `net pvp on` you enter competitive mode: you can hit and be hit, only by those whose grade is close to yours. Picking on those much weaker draws NetGuard. `net hack <nick>` opens the target's network: their personal files, their name and bank credentials, only let themselves be taken after you have been inside for a while. The traces you leave in their network tell them who you are, within an hour.

**From the field:** Go in clean, come out clean: it is the only way not to have a name.

## Theft and the audit

*The ordinary transfer, the line that remains*

With account and password you enter their bank through the front door (Login tab) and make any transfer, up to their balance. The transaction line remains on their statement, in their bank's network: delete it before the daily audit. If the audit finds it, the bank returns the money and NetGuard learns your name. Or you change the line's counterparty to another operator whose credentials you have: the audit accuses them. After the audit the victim's password changes. `net intel` keeps what you have found out about the others.

**From the field:** A successful frame-up is worth more than a theft: but you need two locks open.

## Sieges and races

*Striking back and measuring up*

Whoever finds out who hit them has a week to answer with a siege: `net siege <nick> <stake>`. The target defends the gateway in Security Control; if they hold, they keep the stake, if they fall, they pay a ransom. **Races** (`net races`, `net race join`, `net race run`) put operators of the same tier on the same network: find the finish file and bring it home. The server keeps the time; the top three split the pot.

**From the field:** In a race the winner is not who runs, but who never stops to look around.

## The attack console

*Rehearsal, plan, siege on a bot*

The attacker has a **charge** that refills: each attack costs part of it (scan 1, flood and hijack 2, malware 3, breach and cryptolocker 4). From the player's card, **Rehearse** opens the console against an automatic defender of their grade: you launch with the buttons or the 1-6 keys and see only your own hits, the outlines of the rings and the pressure on the core. **Keep as plan** turns the rehearsal into a plan, which you fine-tune on the timeline (**Plan**) and which goes out with the next siege in place of the automatic waves. Against a server bot you play the siege yourself, live, in the same console.

**From the field:** A good plan does not fire everything at once: it waits for the rings to tire.

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# Microcode

The processor does what its microcode tells it. Change it, and you change the way it works: a Microcode is a card the processor loads to do certain things faster, with less noise or with more staying power, almost always paying for it somewhere else.

### Common, rare, special

*96 numbered cards*

The sixty common ones trade an advantage for a cost of equal weight: faster but noisier, more discreet but slower, one thing at the expense of another. The twenty-four rare ones give more and ask less. The twelve special ones are almost a gift. Each has a number: the Microcode tab in System keeps count of the collection.

**From the field:** A common one well chosen for today's job is worth more than a rare one you do not need.

### Loading

*Slots and rules*

The processor holds from none (up to v3) to five (v19-20). They can only be changed outside missions, from the tab or with `micro on|off <n>`. Two Microcode on the same quantity do not add up: the best advantage and the worst cost are what count. Prepare the set for the job: long intrusion, defence, siege.

**From the field:** Before the contract, look at what you need; then load.

### Finding and losing them

*Dot files, rewards, sieges*

They lie hidden in networks, in files whose names start with a dot: without Inspect you do not see them. Operators' networks hide more of them, and rare and special ones turn up only in big jobs. Online they are won in races and, sometimes, by repelling a siege. If you lose a siege, one of those loaded may get damaged (it yields half until you repair it with `micro repair <n>`), burn out or end up with whoever besieged you.

**From the field:** The best set is worth nothing if the gateway falls.

12

# People

Machines tell the truth about the people who use them. Read enough files and you will know who works where, who they talk to, how much they owe and what they hide. Luck does not count here: what you have found does.

## Dossier

*Four names, five facts*

Every company in the Directory has four employees you can get to know: role, contact, access, debt, secret. Staff and account records give roles, contacts and hosts; accounting ledgers the debts; evidence and hidden files the secrets. One file, one fact. `dossier <company>` or the Dossier button in the Directory.

**From the field:** Before going in, read. Before reading, download.

## Lure

*A message that looks real*

Who sends it, what it is about, where it lands: every true detail makes the message believable. The Black Net needs one, Lumen two, the Concord and NetGuard three. When you have enough, `lure <company> <n>`: a few hours later you have a way in from their host for the next contract. Nobody falls for it twice.

**From the field:** A lure with a wrong fact does not get sent: wait until you know more.

## Insider

*The right lever*

A debt gets paid, a secret gets used. Paid, they stay loyal as long as the fee arrives every week. Blackmailed, they hold for three contracts, and crack at once if you are traced at that company. In return the alert starts lower, one watcher is switched off and every contract brings a tip-off. `insider <company> <n> pay|blackmail`.

**From the field:** Blackmail costs little in credits. You pay the rest later.

## Customer service

*Account recovery*

City, employer, a relative: account holders give themselves away a piece at a time in ledgers and statements. With the answers the bank asks for (two for ordinary people, three for those who matter), `recover <name>` gives you the password.

**From the field:** In defence, the same weapon turns against you: read staff requests carefully before approving them.

13

# Quick reference

| Command | What it does | Program |
|---|---|---|
| scan <node> | reveals defences and files | NetSonar |
| arp <node> | discovers nearby nodes | ArpScan |
| inspect <node> | marks anomalous files | Inspect |
| logparse <node> | follows an operator's chain | LogParser |
| undelete <node> | recovers deleted files | FileUndeleter |
| break <node> [n] | breaks the defences | WallBreaker / ProxyMask / HashRipper / CurveBreaker |
| decrypt <node> <n> | decrypts a file | CipherKey |
| get / rm / edit | download, delete, alter | (node access) |
| db show / alter / delete | works on tables | DBEdit |
| db diag / rebuild | diagnosis and restore | DBDiag / DBRebuild |
| blind / wipe <node> | blinds the monitor / wipes the logs | GhostShell |
| veil | starts or stops the veil | TraceVeil |
| root <node> | permanent relay | DeepRoot |
| corrupt / lock <node> | corrupts / locks | ByteRot / LockJaw |
| bomb <node> / detonate | arms / sets off | TimeBomb |
| devour <node> | destroys the host for good | Devour |
| put <node> <n> / attach | uploads a file / re-downloads the attachment | (node access) |
| revshell <node> | leaves a door open | RevShell |
| keylog <host> / pull <company> | plants / pulls the recorder | KeyWraith |
| lanscan / traverse <node> | locates / crosses | LANScan / LANTraverse |
| relays / relay lease | lists / leases relays | (gateway) |
| oc [0-3] | tunes the processor | (gateway) |
| tasks / cancel <id> | running tasks / cancel | (gateway) |

**TAB** completes commands, node names and file names. `help` lists, at any moment, what the context you are in allows you to do.

## Keyboard shortcuts

| Key | Action |
|---|---|
| F1 / F2 / F3 / F4 | Gateway / Network (in a mission) / Terminal (also Ctrl+T; Applications menu) / System |
| F5 / F6 / F7 / F8 | Black Market (not in a mission) / Messages / Events / Defence Console (in defence and siege) |
| F9 / F10 / F11 | save slots / music on-off / fullscreen |
| Esc | Options; closes modal windows, lists and dialogs; cancels editing a cell |
| Space | pause (in a mission) |
| 1 / 2 / 3 | speed x1 / x2 / x4 (in a mission) |
| Enter | runs the command; accepts a cell edit; confirms dialogs |
| Tab | completes commands, nodes and files; accepts a cell edit |
| Up / Down | command history in the terminal; selection in lists |
| PgUp / PgDn | scroll the terminal output by 10 lines |
| Ctrl+L | clears the terminal screen |

Space and 1/2/3 do nothing while you are typing in the Terminal or in a cell: there the keys are text.

## Mission shortcuts

With the Network window open (and no text field in focus) letters act on the **selected node** exactly like the buttons, with the same checks: if a program is missing or RAM is not enough, the message at the bottom says so. Each button's key appears in its tooltip.

| Key | Action |
|---|---|
| s | scan |
| b | breaks the outermost barrier, with the right program |
| f / p / c / e | breaks the Firewall / Proxy / Password / Elliptic curve barrier |
| m | blinds the monitor (GhostShell) |
| d | downloads a file: the target if it is on the node, otherwise the first one not yet downloaded |
| l | wipes the logs (GhostShell) |
| r | plants a RevShell |
| k / a | KeyWraith / ArpScan |
| n / t | LANScan / LANTraverse |
| i / u | Inspect / FileUndeleter |
| v | starts or stops TraceVeil |
| Tab | selects the next known node |
| Del | cancels the last operation started on the node |

Malware, cryptolocker, rootkit, TimeBomb and removing a RevShell stay on the buttons only: they are heavy actions, and a key pressed by mistake would cost dearly.

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# Habits that save you

| | |
|---|---|
| 1 | Scan before you act, always. A node that looks the same as the last one often is not. |
| 2 | If the node is watched, the monitor gets blinded *before* anything else. |
| 3 | Do not launch everything at once: power is shared, and at the top of the queue there is always something useless. |
| 4 | Take only what the contract needs. The rest is weight, risk and time. |
| 5 | Wipe the logs of the nodes you touched before you leave: every trace is one more hour of attention on you. |
| 6 | Close the RevShells you no longer need. Leaving them open exposes you. |
| 7 | A file that does not change colour is not necessarily clean. Half-trusting is better than trusting completely. |
| 8 | In hard networks someone is watching: if the administrator notices you or the network closes up on a branch, do not insist, change route. |
| 9 | Zero Trace does not exist. Choose it like a budget, not like a hope. |

## Keep the noise down, operator.

*In this Network, whoever makes themselves heard gets found.*